1. Monthly Bill Tracker

Calculate the total of monthly bills. Given a list of bills like [1500, 2000, 3000], iterate through the list. Print each bill amount and the total at the end.

In [54]:
```
# Q1 Solution
total_bills=[1500,2000,30,300,3589,654,22,87]
total=0
counter=1
for bill in total_bills:
    print(f"The Bill amount of bill {counter} is = ₹{bill}")
    total+=bill
    counter+=1
print(f"Total Bill is: ₹{total}")
```

```
The Bill amount of bill 1 is = ₹1500
The Bill amount of bill 2 is = ₹2000
The Bill amount of bill 3 is = ₹30
The Bill amount of bill 4 is = ₹300
The Bill amount of bill 5 is = ₹3589
The Bill amount of bill 6 is = ₹654
The Bill amount of bill 7 is = ₹22
The Bill amount of bill 8 is = ₹87
Total Bill is: ₹8182
```

2. School Attendance

Simulate attendance tracking. Given a list of student names like ["John", "Mary", "Alex"], print each name followed by "Present."

In [55]:
```
# Q2 Solution
students=["John", "Mary", "Alex"]
for student in students:
    print(f"{student} Present.")
```

```
John Present.
Mary Present.
Alex Present.
```

3. Salary Distribution

Distribute bonuses to employees. Given a dictionary {"John": 5000, "Mary": 6000}, iterate through the dictionary. Print each employee's name and their bonus.

In [57]:
```
# Q3 Solution
employees={"John": 5000,"Mary":6000,"Sebastian":7000,"Julie":5000}
for name in employees:
    print(f"{name}'s bonus is = ₹{employees[name]}")
```

```
John's bonus is = ₹5000
Mary's bonus is = ₹6000
Sebastian's bonus is = ₹7000
Julie's bonus is = ₹5000
```

4. Grade Report

Calculate grades for students. Given a list of scores [85, 92, 76, 61], assign grades:

> = 90: A = 80: B = 70: C < 70: F Print the grades for each student.

In [58]:
```
# Q4 Solution
total_marks=[85,92,76,61]
for marks in total_marks:
    if marks>=90:
        print(f"Grade assigned for marks {marks} is A")
    elif marks>=80:
        print(f"Grade assigned for marks {marks} is B")
    elif marks>=70:
        print(f"Grade assigned for marks {marks} is C")
    else:
        print(f"Grade assigned for marks {marks} is F")
```

```
Grade assigned for marks 85 is B
Grade assigned for marks 92 is A
Grade assigned for marks 76 is C
Grade assigned for marks 61 is F
```

5. Book Availability

Check if a book is in a library. Given a list of books like ["Python Basics", "Data Science"], ask the user for a title. Print "Available" if the title is in the list, or "Not Available" otherwise.

In [72]:
```
# Q5 Solution
books_present=["Python Basics","Data Science","Data Analytics","Excel","Basic Python"]
findbook=input("Enter The book's name: ")
found=False
for book in books_present:
    if findbook.lower().replace(" ","")==book.lower().replace(" ",""): #using replace if the user forgets space
        found=True
        break
if found:
    print(f"{findbook} is \"Available\"")
else:
    print(f"{findbook} is \"Not Available\"")
```

```
Excel is "Available"
```

6. Temperature Analysis

Analyze daily temperatures for a week. Given a list [30, 32, 28, 35, 29], find the highest and lowest temperatures using a loop

In [65]:
```
# Q6 Solution
total_temp=[30, 32, 28, 35, 29]
highest_temp=total_temp[0]
lowest_temp=total_temp[0]
for temp in total_temp:
    if temp>highest_temp:
        highest_temp=temp
    if temp<lowest_temp:
        lowest_temp=temp
print(f"Highest temperature is {highest_temp}°C")
print(f"Lowest temperature is {lowest_temp}°C")
```

```
Highest temperature is 35°C
Lowest temperature is 28°C
```

7. Inventory Management

Check for low-stock items in a store. Given a dictionary {"Apples": 5, "Bananas": 20}, print items with quantities below 10.

In [68]:
```
# Q7 Solution
inventory={"Apples": 5, "Bananas": 20,"Guava":18,"Watermelon":7}
print("Items with quantities below 10 units:")
for item in inventory:
```

```
    if inventory[item]<10:
        print(item)
```

```
Items with quantities below 10 units:
Apples
Watermelon
```

8. Flight Seat Booking

Check booked and available seats. Given a list of seats [1, 2, 3, 4] and booked seats [2, 4], print "Booked" or "Available" for each seat.

In [69]:
```
# Q8 Solution
total_seat=[1,2,3,4,5,6,7,8,9,10,11,12]
booked=[2,4,5,9,8]
for seat in total_seat:
    if seat in booked:
        print(f"Seat Number {seat} is Booked")
    else:
        print(f"Seat Number {seat} is Available")
```

```
Seat Number 1 is Available
Seat Number 2 is Booked
Seat Number 3 is Available
Seat Number 4 is Booked
Seat Number 5 is Booked
Seat Number 6 is Available
Seat Number 7 is Available
Seat Number 8 is Booked
Seat Number 9 is Booked
Seat Number 10 is Available
Seat Number 11 is Available
Seat Number 12 is Available
```

9. Savings Interest Calculator

Calculate weekly interest for a savings account. Given a list of daily balances [5000, 5200, 5300], calculate daily interest at 0.05% and display the total interest.

In [71]:
```
# considering every elements present in the list are daily balances of different people
def interest_calculation(balance):
    return (balance*(0.05/100))*7

daily_balance=[5000,5200,5300]
```

```
for balance in daily_balance:
    total_interest=interest_calculation(balance)
    print(f"Total weekly interest for {balance} is {total_interest}")
```

```
Total weekly interest for 5000 is 17.5
Total weekly interest for 5200 is 18.2
Total weekly interest for 5300 is 18.55
```

10. Customer Feedback Analyzer

Analyze feedback for keywords. Given a list of comments like ["Great service", "Poor product"], print comments containing the word "service."

In [52]:
```
# Q10 Solution
total_comment=["Great service", "Poor product","Poor Service","Poor product","Mindblowing service"]
for comment in total_comment:
    if "service" in comment.lower():
        print(comment)
```

```
Great service
Poor Service
Mindblowing service
```

In [ ]: